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भारतीय सूचना प्रौद्योगिकी संस्थान रायचूर
Indian Institute of Information Technology Raichur

NAUKRI GURU: AI-POWERED JOB AUTOMATION PLATFORM

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Guide: Dr. Neha Agarwal

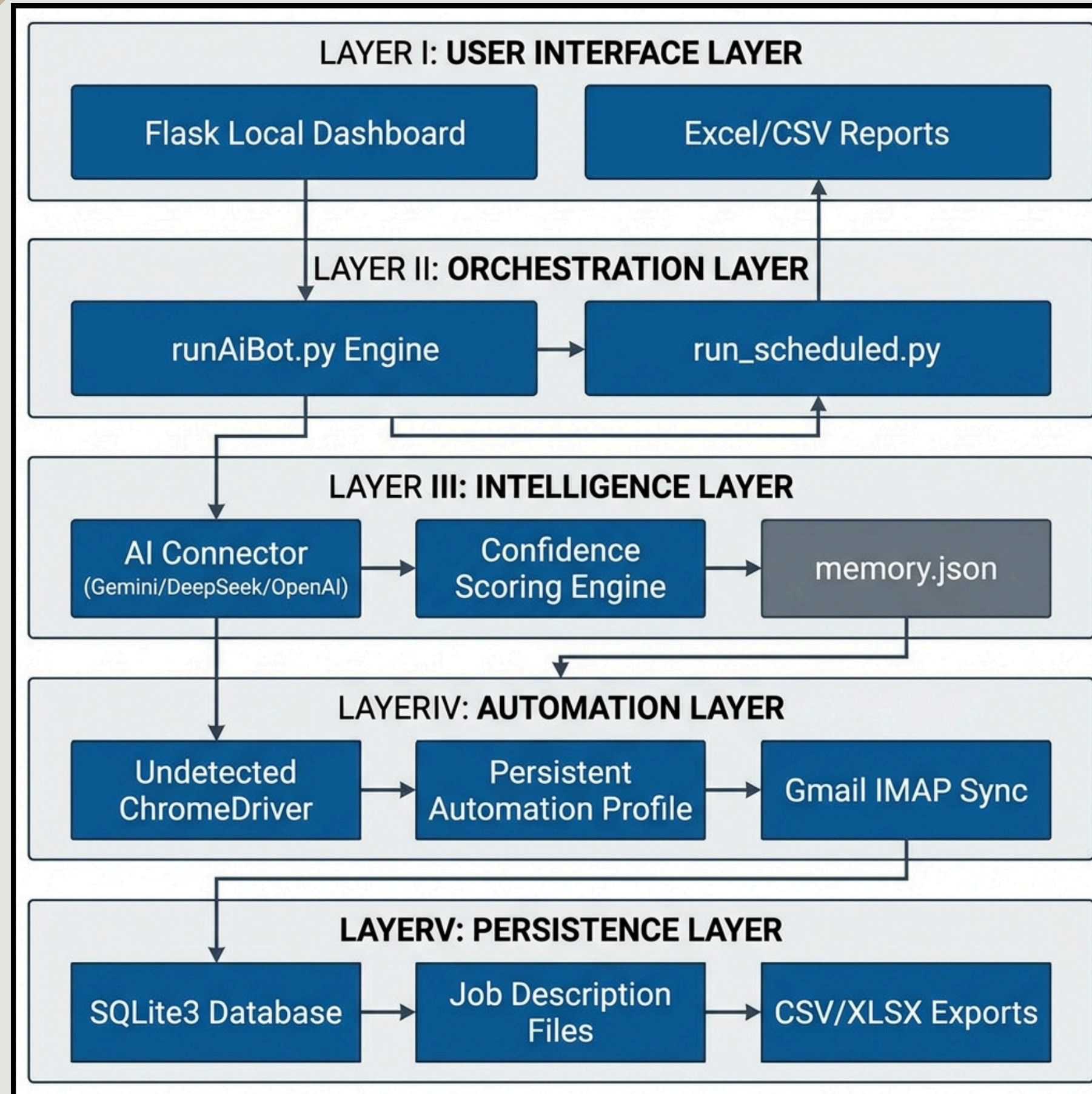
INTRODUCTION

- Job hunting requires hundreds of repetitive applications.
- Candidates waste time answering the same screening questions repeatedly.
- Tracking applications across different platforms is fragmented and manual.

OBJECTIVES OF NAUKRI GURU

- Automate the LinkedIn job application process to save time.
- Score and filter jobs based on profile suitability.
- Answer screening questions autonomously using AI and memory.
- Track application status automatically via Gmail integration.

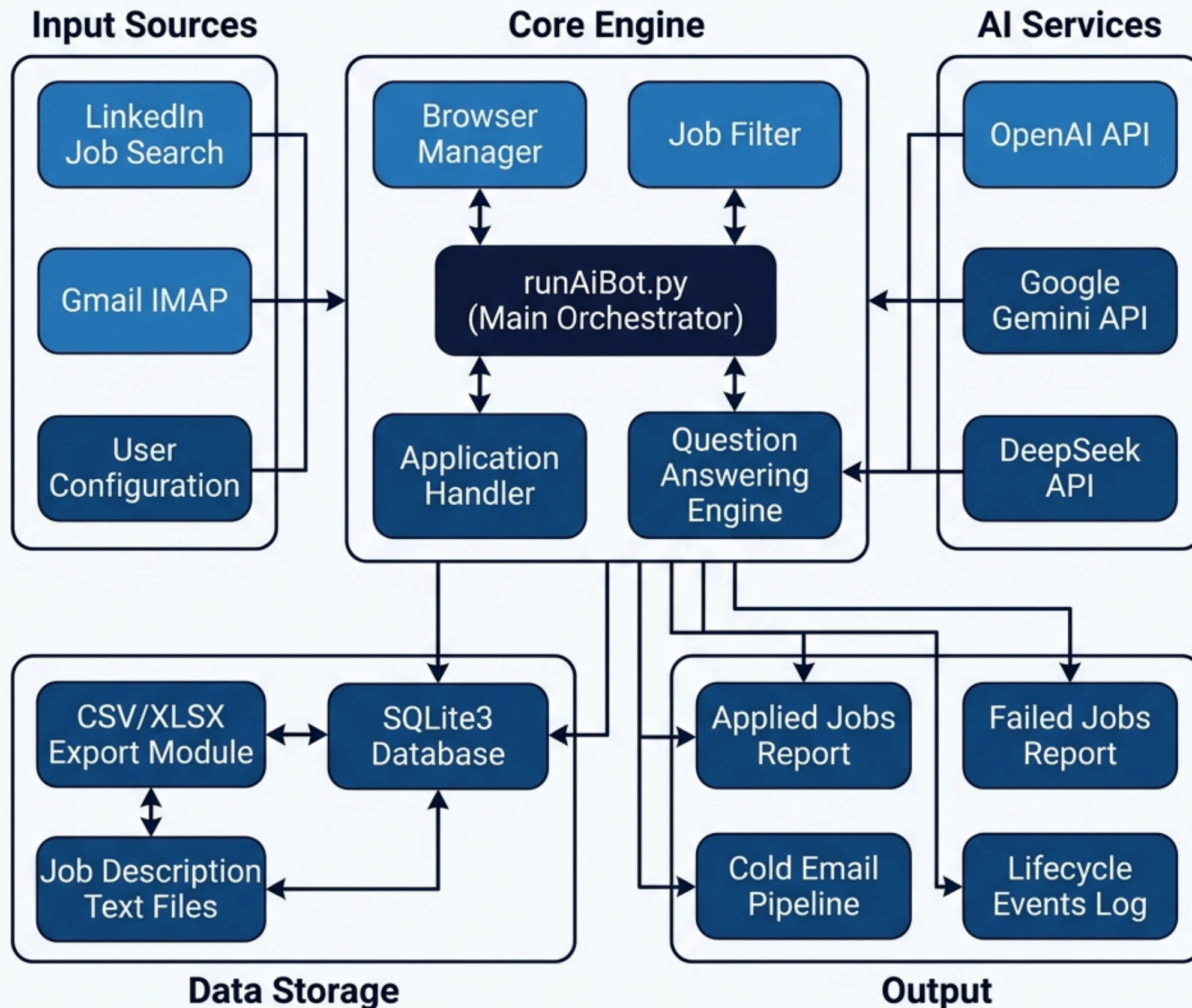
SYSTEM BLOCK DIAGRAM



5-layer modular design

1. **UI**
2. **Orchestration**
3. **Intelligence**
4. **Automation**
5. **Persistence.**

AI-POWERED JOB AUTOMATION PLATFORM ARCHITECTURE

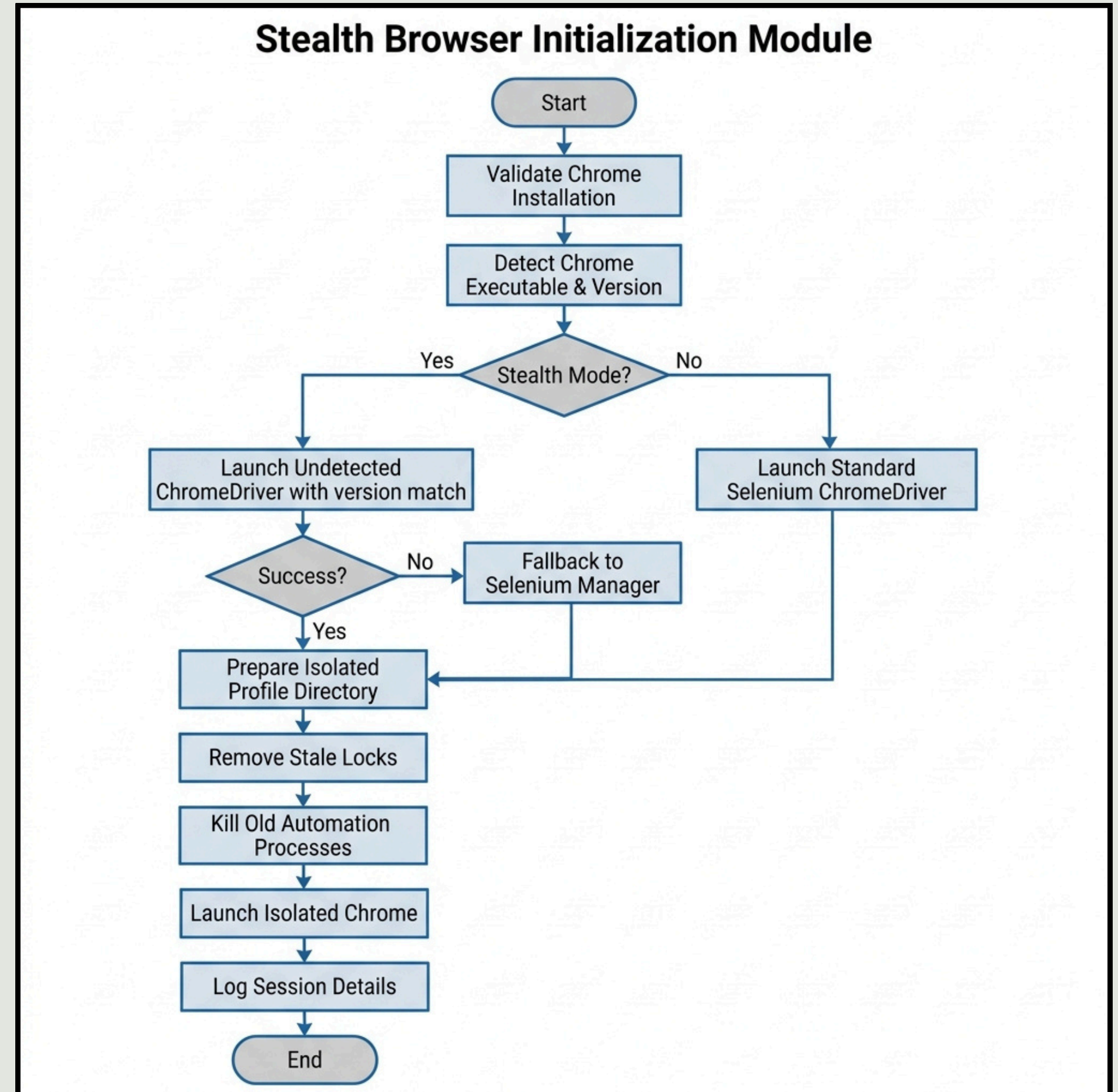


Software Architecture

- **Decoupled Python application for robust execution.**
- **Integrates browser automation, AI, email, and SQLite storage seamlessly.**

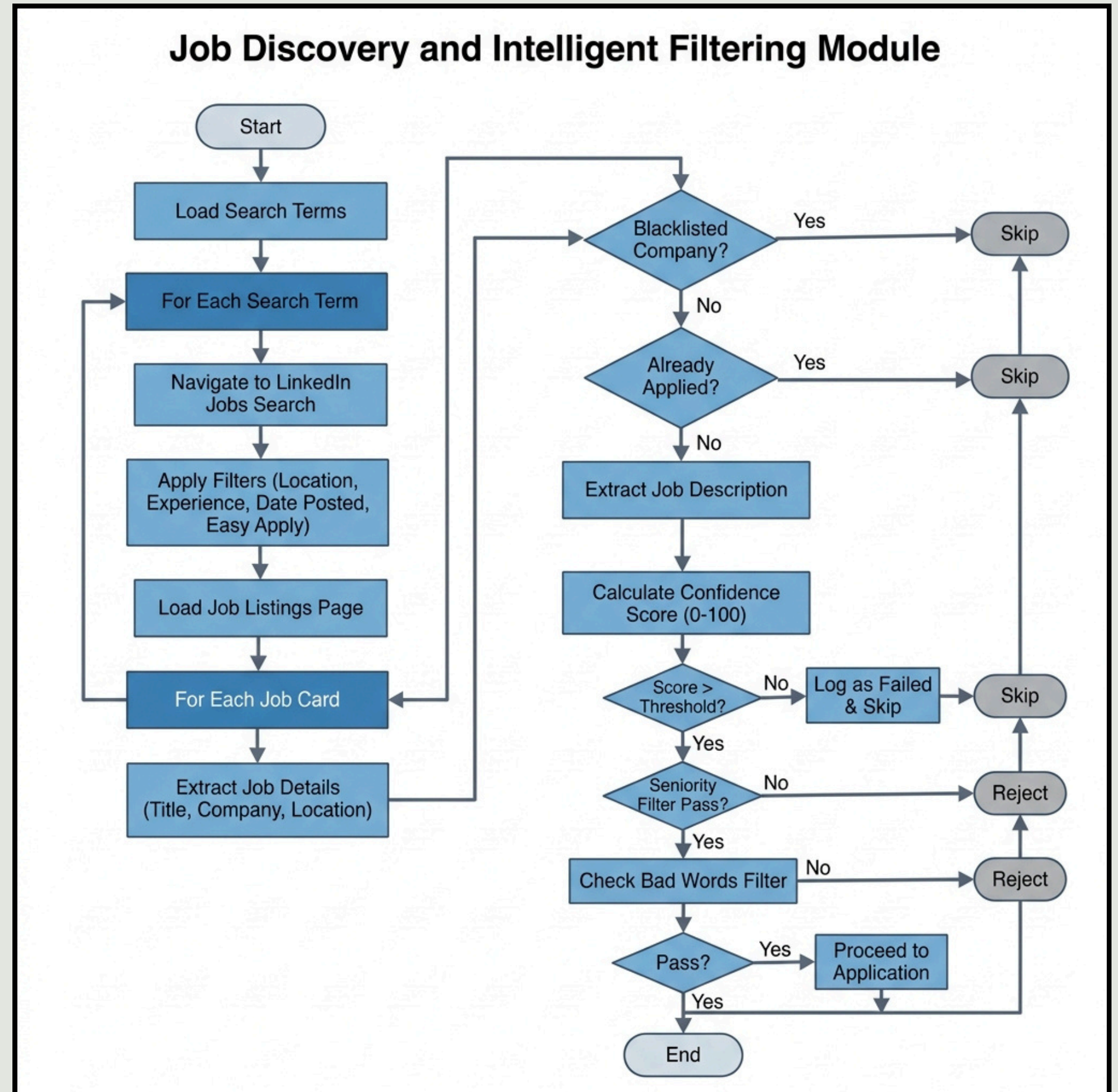
Module 1: Stealth Browser Automation

- Bypasses anti-bot detection using isolated Chrome profiles.
- Auto-detects Chrome versions to prevent startup crashes.



Module 2: Job Discovery & Filtering

- **Automatically searches and extracts metadata from job listings.**
- **Implements a multi-stage funnel to narrow down the candidate pool.**



CURRENT JOB FILTERING CONFIGURATION

Search Target

Location India

Search terms

AI Engineer, ML Engineer

Python Developer Intern

AI / ML Intern

Software Developer / Intern

LinkedIn Filters

Sort Most relevant

Date posted Month, week, 24 hours

Easy Apply False - includes external jobs

Applicants Under 10 disabled

Candidate Fit

Experience Internship, Entry, Associate

Job type Full-time, Internship

Work mode On-site, Remote, Hybrid

Experience cap Current experience: 1 year

Run Controls

Switch term after 5 applications

Max per run 20 applications

Daily auto-run Enabled at 11:00

Gmail sync 20 days, 50 emails

Skip Rules

Company skip word: Crossover

Reject senior or unrelated titles:

Senior, Lead, Architect, Manager

Principal, Director, VP, Consultant

SAP, ERP, Salesforce, Workday

Indeed Scraper

Enabled True

Scrape cap 15 jobs / term

Min salary INR 30k month / 4 LPA

Undisclosed pay Not allowed

HOW THE FILTERING WORKS

Blacklist

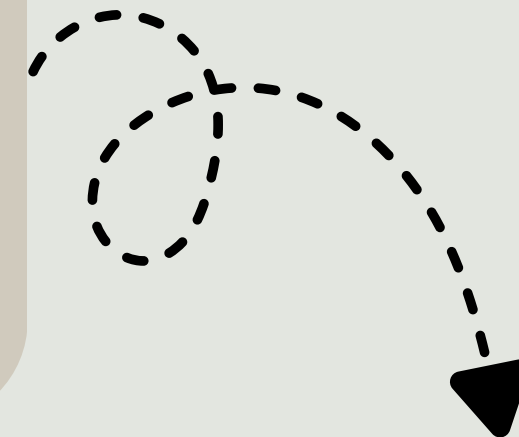
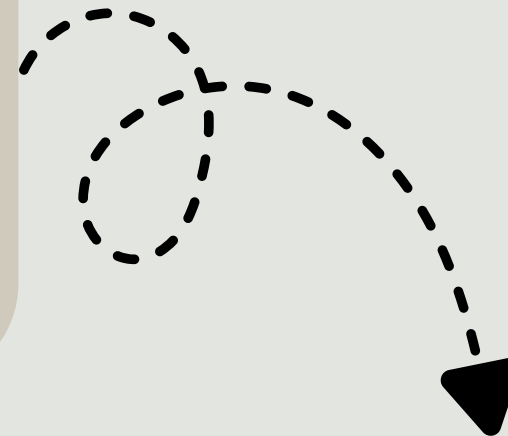
Ignores blacklisted companies and previously applied jobs.

Confidence Scoring

Evaluates Job Description text (+intern, -senior).

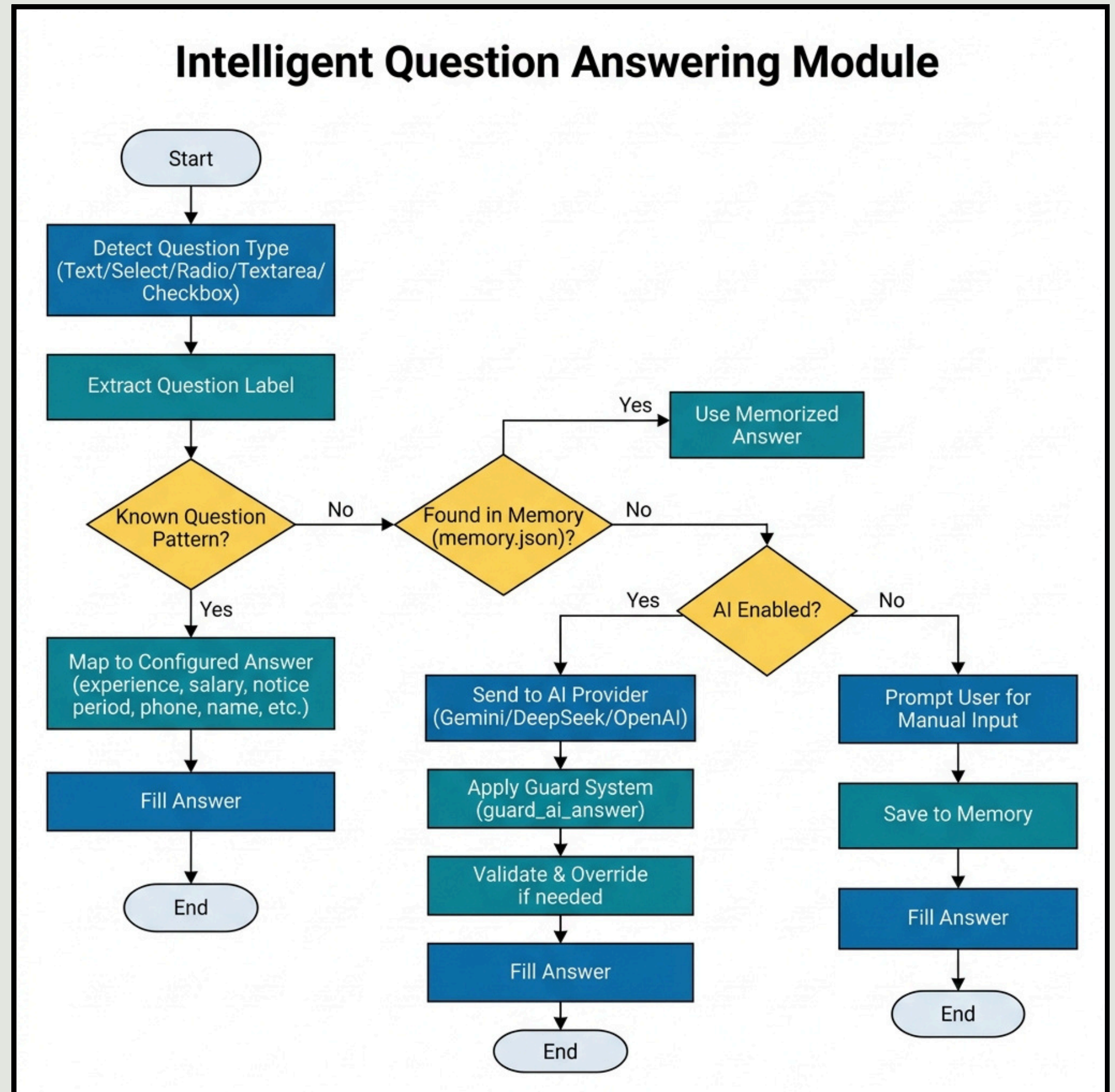
Seniority Check

Bypasses overqualified roles (e.g., Principal, Director).



Module 3: Intelligent Question Answering

- Navigates complex forms (dropdowns, text areas, radio buttons).
- Learns from previous answers to become fully autonomous over time.



THREE-TIER ANSWERING CASCADE

Tier 1 (Config)

Direct mapping
for personal
info (phone,
address).

Tier 2 (Memory)

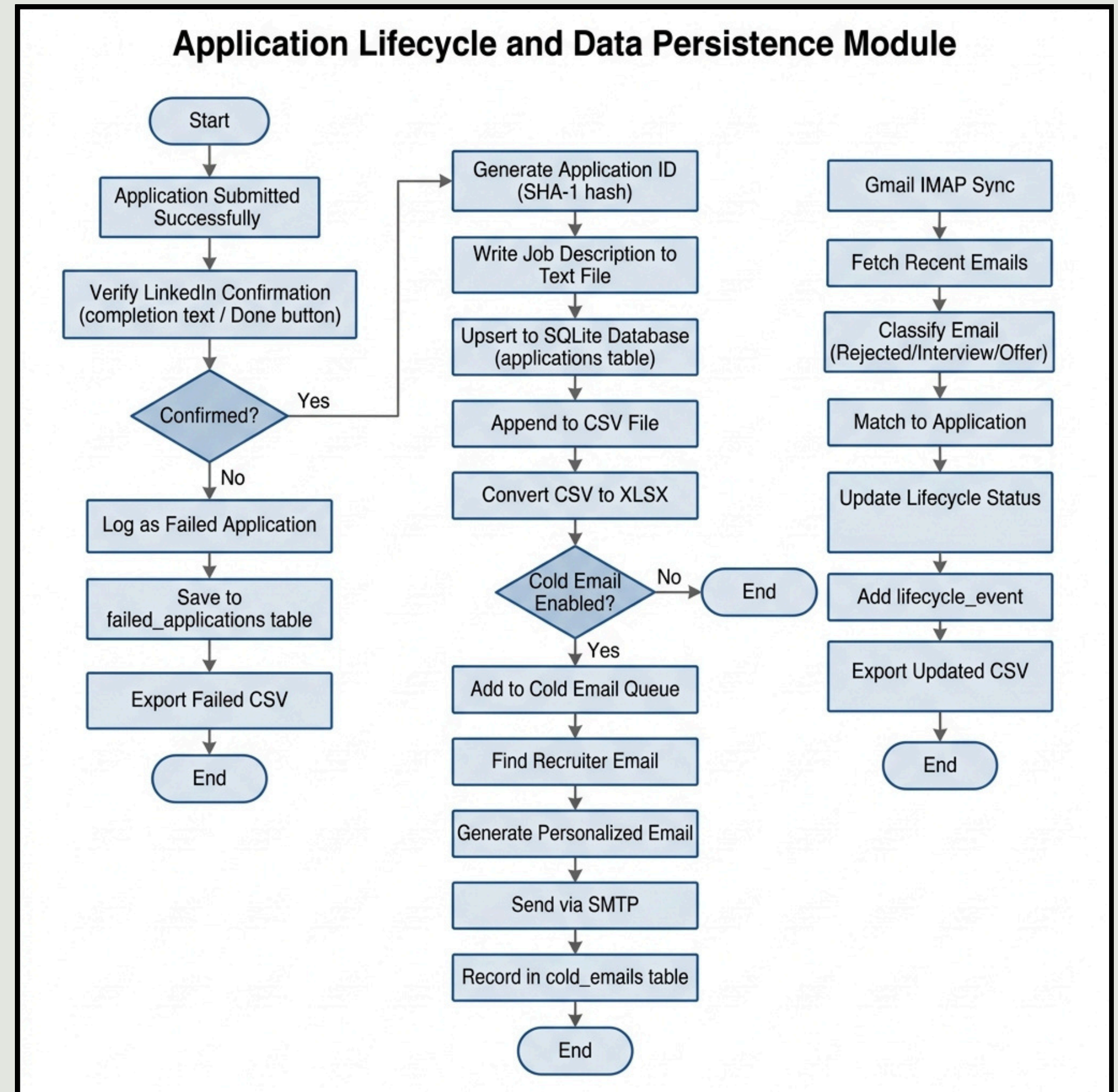
Caches previously
answered
questions for
reuse.

Tier 3 (AI + Guard)

Uses Gemini/OpenAI
for new questions,
protected by a safety
guard to prevent
hallucinations of
sensitive data.

Module 4: Lifecycle & Persistence

- **Verifies successful submission on the screen before saving.**
- **Stores all application data in a local SQLite database.**



TRACKING & RECRUITER OUTREACH

Gmail Sync

Scans inbox to detect recruiter responses and updates application status.

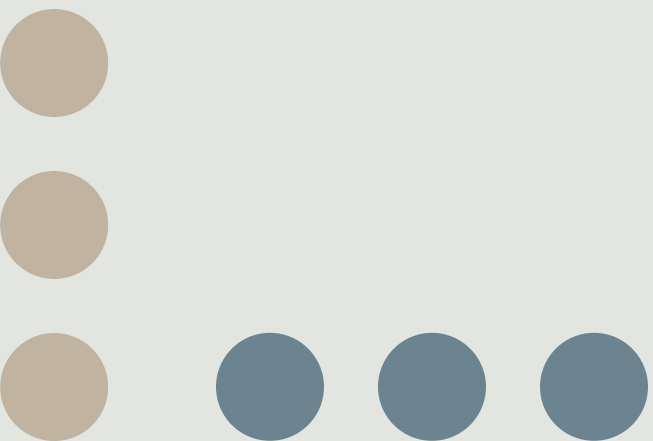
Cold Email

Generates personalized outreach emails to recruiters and safely sends them via SMTP.



MULTI-PORTAL [INDEED] JOB DISCOVERY

- Scrapes Indeed to find a broader range of job opportunities.
- Filters out irrelevant jobs using a bad-word blacklist.
- Saves results in the same way as LinkedIn jobs in a Different tracker.



KEY CHALLENGES SOLVED

- **LinkedIn Changes:** Solved by using a multi-selector cascading strategy.
- **AI Hallucinations:** Solved using safety guard overrides.
- **Duplicate Jobs:** Solved by using a session-level memory cache to avoid reapplying.

RESULTS: STEALTH BROWSER IN ACTION

```
[JOB-DEBUG] Selected job card: Software Engineer | GoogleLocation: Bangalore | Style: Hybrid | Job ID: 3847291056
[SCORE] Apply Confidence Score: 85%
[APPLY-DEBUG] Easy Apply button detected
[EASY-APPLY-MODAL-OPEN] Easy Apply modal detected
[APPLY-CONFIRMED] Completion text detected for job_id=3847291056
[APPLY-SUCCESS] Confirmed application at 2026-05-15 14:23:45
[APPLICATION-MAILED] Successfully saved Software Engineer | Google
---
```

Total runs: 1
Jobs Easy Applied: 5
External job links collected: 2
Total applied or collected: 7
Failed jobs: 1
Irrelevant jobs skipped: 12

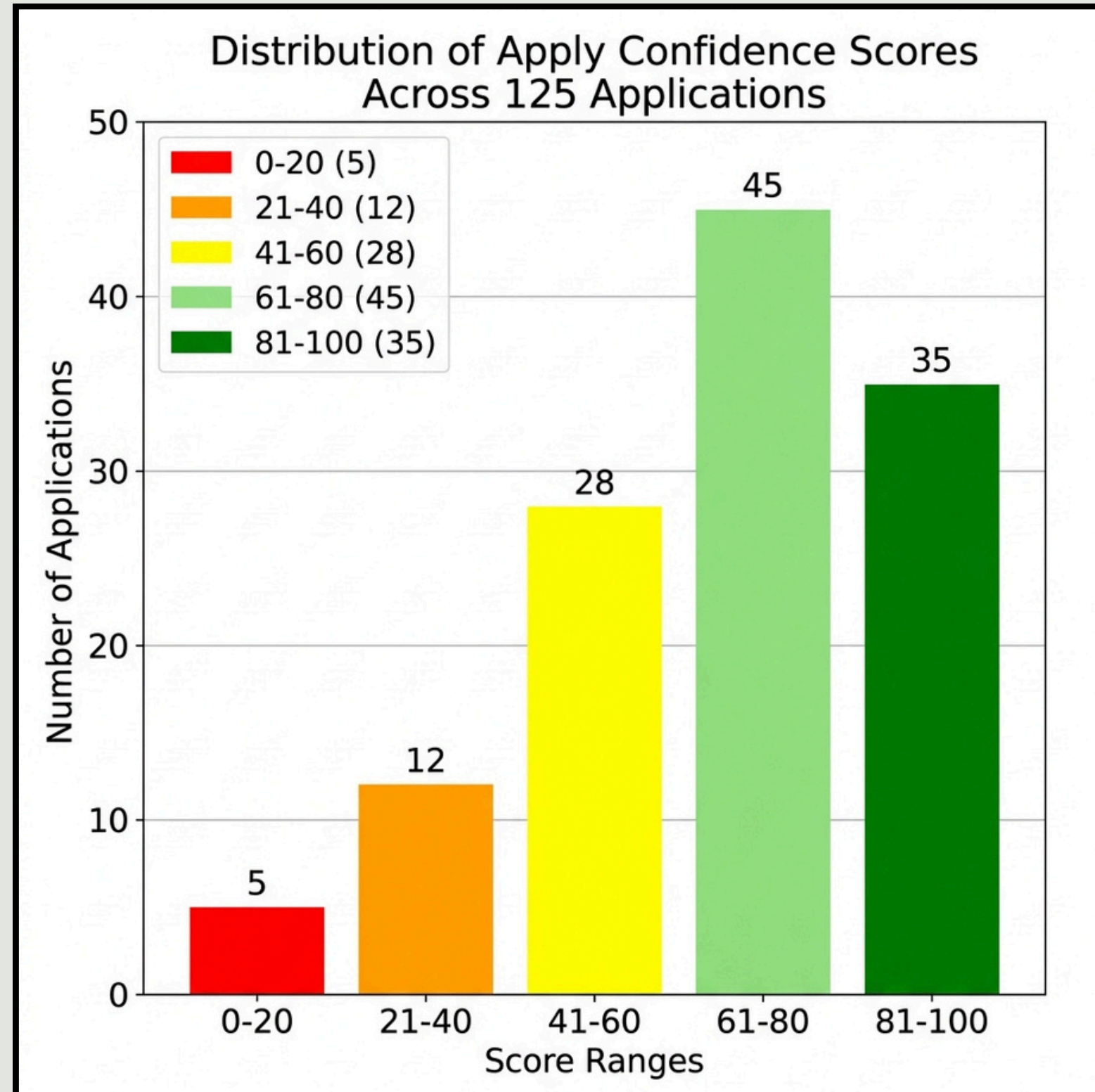
Real-time console tracking of the automation pipeline.

RESULTS: AUTOMATED TRACKING

	Job ID	Title	Company	Location	Work Style	Application Date	Status	Confidence Score	Recruiter Name	Cold Email Status
	3847291056	Software Engineer	Google	Bangalore	Hybrid	2026-05-15	Interview Scheduled	85	Priya Sharma	Sent
	2938475612	Backend Developer	Amazon	Hyderabad	Remote	2026-05-14	Rejected	72	Unknown	Pending
	4829173650	ML Engineer	Microsoft	Pune	On-site	2026-05-13	Applied	90	Rahul Gupta	Sent
	5738291046	Full Stack Dev	Flipkart	Bangalore	Hybrid	2026-05-12	OA Received	78	Ankit Verma	Failed
	6847392051	Data Analyst	Infosys	Chennai	On-site	2026-05-11	Applied	65	Unknown	Pending

- Generates structured XLSX reports.
- Color-coded statuses for easy manual review.

PERFORMANCE METRICS



- Time Savings: ~90% reduction (15-20s per app vs. 3-5 mins manual).
- Success Rate: 87.9% successful submission rate.
- Filtering: Automatically removes ~79% of unsuitable listings.

CURRENT TECH STACK USED



Python
Core language



Selenium
Browser automation



Undetected
Chrome driver startup



SQLite
Local normalized DB



Pandas
CSV/XLSX cleanup



OpenPyXL
Excel exports



Gmail
IMAP sync, SMTP mail



LLM APIs
Gemini, OpenAI, DeepSeek



BeautifulSoup
HTML parsing



Requests
HTTP helpers



PyAutoGUI
Local prompts



Flask
Local dashboard/API

PERFORMANCE METRICS

Can be added for scale



LangChain

AI chains, agents, tools



LlamaIndex

Resume/JD retrieval



Celery Queue

Background workers



Redis

Cache and queue broker



PostgreSQL

Concurrent production DB



Docker

Repeatable deployment



FastAPI

Service/API layer



Playwright

Stable browser tracing



Vector DB

Semantic job matching



Kubernetes

Orchestrate workers



Monitoring

Sentry, Grafana, metrics

Why not used here?

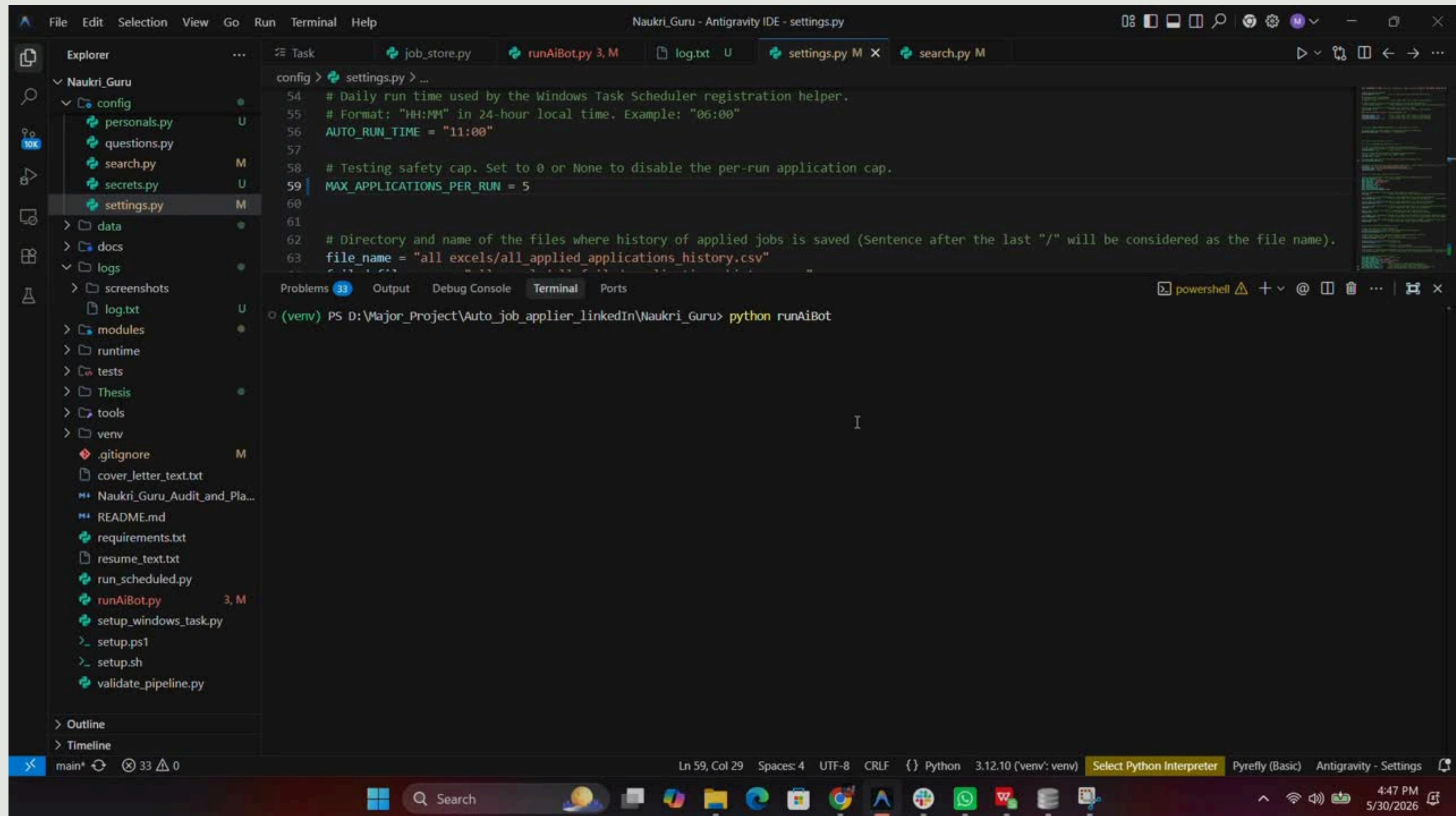
Single-user local prototype

SQLite and files are enough

Lower setup complexity

Direct Selenium is simpler to debug

NAUKRI_GURU IN ACTION



CONCLUSION & FUTURE SCOPE

Conclusion

Successfully built an autonomous, self-improving job application pipeline.

Future Scope

Expanding auto-apply to platforms like Glassdoor and adding a web-based analytics dashboard.



Thank You

For your attention